

Muhammad Afin Atsal

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Summary

Fresh graduate in Information Technology with hands-on experience building real-world AI and software solutions, from computer vision systems to conversational applications, through cross-functional teamwork in fast-paced, Agile environments. Comfortable turning open-ended problems into working products, learning new tools quickly, and communicating technical ideas clearly to both technical and non-technical audiences. Experienced as a Python instructor and mentor, guiding students through hands-on technical projects from concept to completion.

Education

University of Brawijaya

August 2022 - July 2026

Bachelor of Information Technology

- GPA 3.67/4.00

- **Thesis:** "Multi-Stage Waste Detection Model Based on YOLO and Hierarchical Multi-Label Classification" developed a two-stage pipeline (YOLOv12 + HSCN) achieving 0.84 F1-score (detection) and 0.97 mAP (classification)
- **Related Coursework:** Machine Learning, Neural Networks, Computer Vision, Large Language Models (LLMs), Artificial Intelligence, Data Mining, Big Data Analytics, Data Structures & Algorithms, Data Visualization, Database Systems, Database Programming, Software Engineering, DevOps Engineering

Professional Experience

X-Camp, PT XLSmart Telecom Sejahtera Tbk

Aug 2025 - Dec 2025

AI/ML Engineer (MBKM Program)

Jakarta, Indonesia

- Collaborated with a cross-functional team under X-Camp's Open Lab to design and ship 3 AI/IoT solutions, following Agile-Scrum methodology with weekly sprints and stakeholder demos.
- Built a computer vision-based Visual Inspection system using CNN to automate defect detection on production lines, achieving 99.6% accuracy and 99.6% mAP@50, then integrated it into a real-time ThingsBoard dashboard for factory-floor monitoring.
- Designed an Object Removal Detection system using OpenCV template matching to solve a real safety-monitoring problem, translating the client's operational needs into a working alerting system.
- Developed a real-time People Counting System using YOLOv12l and BoT-SORT tracking, working closely with the team to define zone logic and validate results against real video footage.
- Built a RAG-based internal chatbot using FastAPI and Gemini Flash 2.5, presenting the solution to non-technical stakeholders to gather feedback and iterate.

IT Department, PT Amman Mineral Internasional Tbk

Jul 2025 - Aug 2025

Back-End & AI System Developer (Internship)

Sumbawa Barat, Indonesia

- Built the back-end and AI system for an interactive RAG-based chatbot to automate boat ticket booking within the company's internal SuperApps ecosystem.
- Designed a modular microservices architecture using FastAPI, integrating real-time scheduling data with PostgreSQL + PGVector for semantic search and retrieval.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with Google Gemini API, supporting bilingual (Indonesian/English) multi-turn conversations with contextual memory.
- Achieved average API latency below 2 seconds post-initial embedding; validated system through Postman API testing and Flutter front-end integration.

Teaching Experience

Fakultas Ilmu Komputer (FILKOM) & Fakultas MIPA, Universitas Brawijaya

Feb 2026 - Jun 2026

Python Practicum Instructor (Joint Program)

Malang, Indonesia

- Selected as one of the practicum instructors in a cross-faculty collaboration between FILKOM and FMIPA, teaching Python programming to students for one semester.
- Co-developed curriculum and practicum materials alongside instructors from both departments.
- Evaluated student assignments and provided technical mentoring throughout the semester.

Freelance
Private Tutor, IoT and Applied AI Systems

[periode]
Malang, Indonesia

- Guided high school students from problem definition to working prototype, mentoring them in building an E-Nose system for their scientific research competition.
- Taught Python programming and embedded systems development, breaking down complex technical concepts into approachable lessons.
- Supported the full product journey: sensor data acquisition, preprocessing, and AI-based classification, while encouraging students to think critically and solve problems independently.

Organizational and Event Experience

KBMSDI FILKOM UB
Deputy Head of Entrepreneurship Department

Feb 2024 - Dec 2024
Malang, Indonesia

- Successfully executed 4 main programs, including a seminar and business plan competition aimed at fostering student entrepreneurship.
- Initiated and managed the DSI Store program, producing and selling merchandise (e.g., t-shirts, stickers), generating profit to support organizational funding.
- Led the design and production of official PDH (Organizational Jacket) for KBMSDI members, enhancing organizational identity and unity.
- Created and managed content for entrepreneurial-themed social media posts, supporting brand visibility and knowledge sharing.
- Actively contributed to strategic planning to improve department performance and organizational sustainability.

KBMSDI FILKOM UB
Staff of Entrepreneurship Department

Feb 2023 - Dec 2023
Malang, Indonesia

- Contributed to the DSI Store merchandise production and sales, supporting organizational revenue generation.
- Assisted in planning and executing entrepreneurial events and contributed to internal coordination for smooth program delivery.

FILAFEST (Filkom Anniversary Festival)
Head of Sponsorship and Fundraising

Apr 2023 - Dec 2023
Malang, Indonesia

- Developed and led sponsorship acquisition strategies, successfully securing partnerships with multiple external stakeholders.
- Managed end-to-end proposal development, negotiation, and reporting, ensuring professional communication and mutual value delivery with sponsors.
- Collaborated with other divisions to allocate funds efficiently across events and support the overall success of FILAFEST 2023.
- Raised funds exceeding initial targets, contributing significantly to the festival’s operational budget.

Certifications

• Building AI Agents and Agentic Workflows Specialization - IBM	2026
• Machine Learning Specialization - DeepLearning.AI & Stanford Online	2026
• Google AI Professional Certificate - Google	2026
• Google AI Essentials Specialization - Google	2026

Skills

- **Hard skills:** Machine Learning (Advanced), Computer Vision (Advanced), Deep Learning (Advanced), Natural Language Processing & LLMs (Intermediate-Advanced), Retrieval-Augmented Generation (RAG) (Advanced), Data Analysis (Intermediate-Advanced), Backend Development (Intermediate), ETL & Data Preprocessing (Intermediate), Data Visualization (Advanced), Statistical Testing.
- **Soft skills:** Leadership, Critical Thinking, Problem-Solving, Adaptability, Teamwork and Collaboration, Time Management, Project Coordination, Analytical Thinking, Communication & Technical Mentoring.
- **Tools and Tech stack:** Python, PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, YOLO, LangChain, FastAPI, Gradio, PostgreSQL, PGVector, Docker, Flutter, Firebase, Git, GitHub, Pandas,

NumPy, Matplotlib, Seaborn, Plotly, Power BI, SQL, Apache Spark, Hadoop, ThingsBoard, VS Code, Jupyter Notebook, Google Colab, Swift, SwiftUI.